Lecture 4: Maximum Likelihood Estimation (MLE)

Intro to statistics notation

- p(event) is the probability of that event
  - p(fair die lands on 1) = 1/6
  - p(fair coin lands on heads) = 1/2
  - p(two fair coins both land on heads) = 1/4
- p(event|fact) is the probability of the event ***given*** the fact
  - p(2 fair dice equal 12|first die is 1) = 0
  - p(fair coin lands on heads|it landed on heads last time) = 1/2
  - p(two fair coins both land on heads|the first coin landed on heads) = ½
- E(event) is the *expected value* of that event

Experiment

- You buy a magical coin and want to know the probability of heads (or tails, for that matter) in the hopes that you can win coin flips more than 50% of the time. How do you figure it out?
  - Flip it a few times:

    Heads, heads, tails, heads, tails
  - What is the probability that a given flip will be heads?

    Well, 3/5 or 60% based on what we've observed
  - Well, why is this?

Experiment - data

- Sequence of events, D = (HHTHT…), k heads out of n flips
- Flips are *independent* and *identically distributed* (i.i.d.)
  - This means, $p(HHTHT) = p(H) * p(H) * p(T) * p(H) * p(T)$
- Hypothesis is that $p(H) = \theta$, therefore $p(T) = 1-\theta$
  - $p(HHTHT|\theta) = \theta^k * (1-\theta)^{n-k}$
- Generally, $p(D|\theta) = \theta^k(1-\theta)^{n-k}$

Experiment – **M**aximum **L**ikelihood **E**stimation (MLE)

- Maximum likelihood estimation: select θ such that it *maximizes* the probability (*likelihood*) of your observed data
  - Bad example: Say probability of heads, p(H), or θ is 1
  - This means that the probability of tails, p(T), or θ-1 is 0
  - So p(HHTHT) would be 0
  - Obviously, the probability of your *observed* data shouldn't be 0%…

Experiment – **M**aximum **L**ikelihood **E**stimation (MLE)

- Maximum likelihood estimation: select θ such that it *maximizes* the probability (*likelihood*) of your observed data
  - θ_{MLE} = arg max(θ) [p(D|θ)]
  - θ_{MLE} = arg max(θ) = [log[p(D|θ)]]
  - θ_{MLE} = arg max(θ) = [log[$\theta^k(1-\theta)^{n-k}$]]

- Set the derivative to zero:

$$\frac{d}{d\theta} \log P(\mathcal{D}|\theta) = 0$$

Experiment – **M**aximum **L**ikelihood **E**stimation (MLE)

- Maximum likelihood estimation: select θ such that it *maximizes* the probability (*likelihood*) of your observed data

  $d/d\theta[\log[p(D|\theta)] = 0$

  $d/d\theta[k*\log\theta + (n-k)*\log(1-\theta)] = 0$

  $k/\theta - (n-k)/(1-\theta) = 0$

  $k/\theta = (n-k)/(1-\theta)$

  $k*(1-\theta) = \theta*(n-k)$

  $k*(1-\theta) - \theta*(n-k) = 0$

  $k - \theta n = 0$

  $\theta_{MLE} = k/n$

Experiments – **M**aximum **L**ikelihood **E**stimation (MLE)

- $\theta_{MLE} = k/n$
- Two experiments:
  - Flip a coin 5 times, get 3 heads
  - Flip a coin 50 times, get 20 heads
- Results:
  - First experiment, $\theta_{MLE} = k/n = 3/5$ or 60%
  - Second experiment, $\theta_{MLE} = k/n = 20/50$ or 40%

- Which is right? Why?

MLE – more is better, but how much is needed?

- $\theta_{MLE} = k/n$
- If you go to infinity (flip the coin infinite times):

  $E(\theta_{MLE})$ = actual, real θ, which we'll call θ^*

- Hoeffding's inequality says that for an epsilon, $\varepsilon > 0$:

  $p(abs[\theta_{MLE} - \theta^*] \geq \varepsilon) \leq 2e^{-2n\varepsilon^*\varepsilon}$

  where n is the number of trials (coin flips in this example)

  As n increases, the right hand side gets very small (close to zero) and the probability that you're close to the true θ, θ^*, gets very close to one

Example with θ^*

- PAC (Probability Approximate Correct)
- Example: I want to know θ^* within $\varepsilon = 0.1$, with probability of at least $1-\delta = 0.95$ (probability within at least 95%). How many coin flips are needed for this?

$2e^{-2n\varepsilon^*\varepsilon} = \delta$

$\varepsilon = \text{sqrt}(\log(2/\delta)/2n)$

$\text{abs}(\theta_{MLE} - \theta^*) \leq \text{sqrt}(\log(2/\delta)/2n) = 0.1$

$\log(2/\delta)/2n = 0.01$

$\log(2/\delta) = 0.02n$

$n = \log(2/\delta)/0.02$

What about for continuous variables?

- What if you don’t have “heads” and “tails,” but 0.1234, 198.286, etc.?
- Gaussian model

$$P(x \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

Properties of Gaussians

- Affine transformation:
  - If $X \sim N(\mu, \sigma^2)$
  - $Y = aX + b \rightarrow Y \sim N(a\mu + b, a^2\sigma^2)$

- Sum of Gaussians:
  - $X \sim N(\mu_x, \sigma_x^2)$
  - $Y \sim N(\mu_y, \sigma_y^2)$
  - $Z = X + Y \rightarrow Z \sim N(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)$